

Jonathan Ma

801-673-1386 | jonathanma217@outlook.com

LinkedIn | GitHub | Personal Website

Education

M.S.E. Applied Mathematics and Statistics (In Progress)

Aug 2025 – Dec 2026

Johns Hopkins University

Baltimore, MD

- Coursework: Bayesian Statistics, Nonlinear Optimization, Probabilistic Machine Learning, Nonparametric Bayesian Statistics, Time Series Analysis, AI and Statistical Methods for Clinical Trial Design and Analysis

B.S. Finance, B.A. Economics, Minor in Statistics

Aug 2021 – May 2025

Rutgers University

Camden, NJ

- Coursework: Econometrics, Applied Data Mining and Machine Learning, Regression Analysis and Time Series, Applied Statistics, Statistical Models, Data Visualization, Textual Analysis in Finance, Machine Learning for Business Applications
- Honors: Aetheneum Honors Society, Honors in Economics, Summa Cum Laude

Research Experience

Research Assistant, Dr. Tengfei Zhang (Finance)

Oct 2023 – July 2025

Rutgers University

- Built a large-scale text analysis pipeline by scraping 7,500+ earnings call and conference transcripts from Capital IQ using Python, followed by data cleaning and preprocessing in R
- Developed a sentiment analysis framework combining GPT-based prompting with structured financial interpretation; manually validated outputs against analyst sentiment classifications (neutral, bullish, bearish)
- Constructed and curated a structured dataset of equity analyst profiles by cross-referencing Capital IQ records with external sources to ensure data completeness and accuracy

Research Assistant, Dr. Tetsuji Yamada (Health Economics)

Jan 2023 – Dec 2025

Rutgers University

- Synthesized findings from 60+ academic studies to inform model specification for adolescent risk behavior analysis
- Developed and implemented a structural equation modeling (SEM) framework on cleaned YRBS data to analyze relationships between behavioral, demographic, and socioeconomic factors

Research Assistant, Dr. Emara Noha (Labor Economics)

Jan 2023 – May 2023

Rutgers University

- Synthesized empirical results and supporting literature to draft a research manuscript and develop conference presentation materials on telecommunications in Egypt

Teaching & Academic Experience

Teaching Assistant, Machine Learning Applications in Business

Aug 2024 – Dec 2024

Rutgers University

- Designed and graded assignments and solutions covering learning algorithms and data fitting
- Supervised Zoom meeting and dealt with technical issues, took attendance, and optimized the transcription
- Held office hours to help students who need help with coding models or getting Anaconda/Github to work

Teaching Assistant, Bayesian Statistics for Physical Sciences
Johns Hopkins University

Jan 2026 – May 2026

- Designed and graded assignments and solutions covering Bayesian inference, probabilistic modeling, and decision theory
- Developed additional conceptual and computational exercises to reinforce key topics, including MCMC and model-based inference
- Produced structured lecture notes and implemented computational examples using Python and Mathematica to support instruction

Statistics & Calculus Tutor
Rutgers University

Jan 2022 – May 2025

- Delivered 150+ tutoring sessions in statistics and calculus, translating complex quantitative concepts into accessible explanations
- Designed and led supplemental instruction sessions, contributing to a 15% increase in average exam scores
- Mentored students on coursework, problem-solving strategies, and academic planning across diverse disciplines

Club Leadership
Rutgers University

Oct 2022 – May 2025

- Advised leadership strategy, resulting in 20+ events and 400% membership growth
- Built and maintained an organizational database and internal handbook (“The Big Book of SFA”) to support leadership continuity and operations
- Designed and executed targeted outreach campaigns, optimizing flyer placement and messaging to maximize student engagement

Organizations

Beta Alpha Psi, Beta Gamma Sigma, Omicron Delta Epsilon, Philadelphia Council of Business Economics

Projects

Stock Forecast Dashboard

GitHub

- Built an end-to-end neural time series forecasting system using N-BEATS and N-HiTS architectures in PyTorch Lightning on financial data retrieved via the yfinance API
- Designed a modular data pipeline for ingestion, preprocessing, and exploratory analysis, including handling missing data, constructing time-indexed series, and generating technical indicators
- Implemented rolling-window validation and out-of-sample evaluation to simulate real-world forecasting performance and avoid lookahead bias
- Conducted systematic hyperparameter tuning (e.g., input window size, forecast horizon, network depth) to optimize predictive performance
- Benchmarked neural models against ARIMA and LSTM baselines, achieving 18–25% reduction in RMSE across test periods
- Developed an interactive dashboard using Plotly/Dash to visualize historical data, model predictions, and forecast uncertainty in real time

Research Topic Lineages (Dynamic Topic Modeling + Diffusion)

GitHub

- Built an end-to-end pipeline to model temporal evolution of biomedical research topics using PubMed data, including ingestion, preprocessing, and storage in SQLite
- Generated document embeddings using transformer-based models and performed topic discovery via HDBSCAN clustering on a per-period basis
- Developed a dynamic topic alignment framework to construct topic trajectories across time, enabling analysis of persistence, emergence, and disappearance of research themes
- Modeled semantic drift in embedding space using PCA-based trajectory analysis, quantifying topic movement and stability via cosine similarity and movement norms
- Implemented a diffusion-based generative model (PyTorch) to learn stochastic dynamics of topic evolution and simulate plausible future trajectories under uncertainty
- Designed visualization tools (Plotly/Matplotlib dashboards) to analyze lineage structure, topic movement distributions, and high-dimensional trajectory projections

innovcal (Probabilistic Forecast Calibration under Misspecification)

GitHub

- Developing a probabilistic forecasting and uncertainty quantification framework for studying forecast calibration under innovation misspecification in stochastic time-series systems
- Implemented diffusion-based innovation modeling by learning flexible residual distributions from forecasting errors using denoising diffusion and score-based generative methods
- Built recursive probabilistic forecasting pipelines that propagate learned innovation samples through VAR and DeepAR forecasting systems to generate predictive trajectories and uncertainty distributions
- Integrated Wasserstein-based perturbation analysis to evaluate forecast robustness, stress scenarios, and calibration degradation under controlled distributional shifts
- Designed controlled stochastic simulation environments with Gaussian, heavy-tailed, skewed, and heteroskedastic innovation processes for evaluating probabilistic forecast behavior under misspecification
- Developed modular evaluation tools for probabilistic calibration and uncertainty diagnostics, including PIT analysis, Expected Calibration Error (ECE), CRPS, energy scores, and predictive interval coverage
- Built reproducible experimentation and logging workflows for comparing Gaussian, bootstrap, Student- t , and diffusion-based innovation models across forecasting environments

Personal Website AI Assistant | Demo

GitHub

- Developed a full-stack retrieval-augmented generation (RAG) system that answers questions about research, projects, coursework, and technical experience using semantically retrieved source documents
- Engineered a modular AI assistant architecture with a Next.js/TypeScript frontend and FastAPI backend supporting document ingestion, chunking, embedding-based retrieval, prompt orchestration, and grounded LLM responses
- Implemented scalable semantic search infrastructure using Supabase vector storage and metadata-aware retrieval pipelines to support extensible knowledge base expansion and retrieval-quality improvements
- Deployed and maintained a public-facing AI portfolio assistant using Vercel and Railway, incorporating API rate limiting, session controls, and responsive conversational UI/UX features
- Designed the system for future extensibility including source-aware retrieval, corpus scaling, retrieval filtering, and integration of domain-specific research copilots

Publications

Bayesian Hierarchical Modeling and Clustering for Malignant Cancer Diagnosis 2026
Argumentation Based Systems Journal

- Developed a 3-level Bayesian hierarchical logistic regression model (tumor, patient, region) to analyze late-stage cancer diagnosis using SEER data
- Performed full Bayesian inference using NUTS with convergence diagnostics (ESS, $\hat{R}$) and posterior predictive validation
- Quantified variance decomposition via intra-class correlations, showing dominant patient-level heterogeneity (ICC = 0.94) relative to region-level effects
- Applied Gaussian Mixture Models to posterior patient-level random effects, identifying latent risk clusters with distinct demographic and tumor profiles

Analyzing the Obesity of Children and Adolescents and the Effect of the Pandemic 2025
Celebration of Undergraduate Research and Creative Activity

- Analyzed adolescent obesity determinants using pooled NSCH data ($n \approx 39,700$) within the PRECEDE–PROCEED causal framework
- Estimated logistic regression models with average marginal effects to quantify impacts of socioeconomic status, healthcare access, and demographic factors
- Measured health inequality using Concentration and Kakwani indices, revealing disproportionate burden among lower-income and minority populations
- Built reproducible data pipelines in R/SQL, incorporating survey weighting, bootstrap inference, and diagnostic checks (VIF, reliability metrics)

Technical Skills

Programming: Python (PyTorch, nltk, MLflow, statsmodels, Numba, polars, PostgreSQL, bs4, arviz), R (lme4, dplyr, reticulate, ineq, CARET, BART, lavaan), SQL, Julia, C++, JavaScript, MATLAB, Mathematica

Statistical Methods: Bayesian Computation, Multiple Time Series Analysis, Mixed Effects Models, Structural Equation Models, causal inference, nonparametric inference, diffusion models, Clustering Algorithms

Tools: STATA, SAS, LaTeX, Tableau, Weka, Jamovi, Microsoft Office Suite, Jupyter, Google Colab, Bloomberg Terminal

Certifications

CITI Good Clinical Practice (2026)

CITI Biomedical Research with Human Subjects (2026)

CFI Financial Modeling and Valuation Analyst (2024)

Bloomberg Market Concepts (2022)

Google Data Analytics (2023)

Columbia+ Machine Learning 1 (2025)

Research Interests

Conformal Prediction and Calibration, Efficient Bayesian Inference, Macroeconometrics, Data-Driven Optimization, Scaling Laws for LLMs, Fair Agentic Systems, Causal Inference, Dimensionality Reduction, Stochastic Generative Modeling